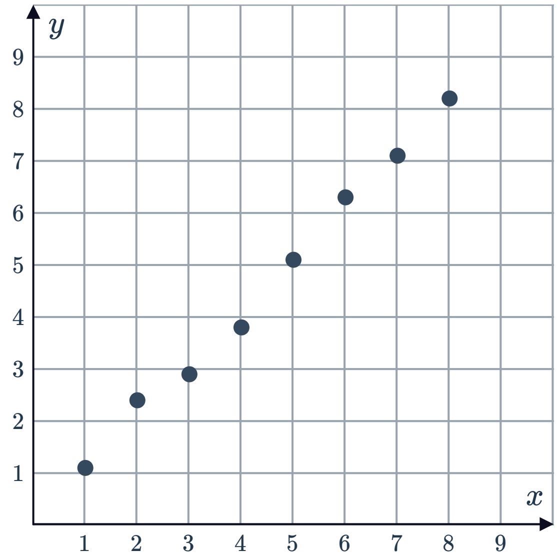
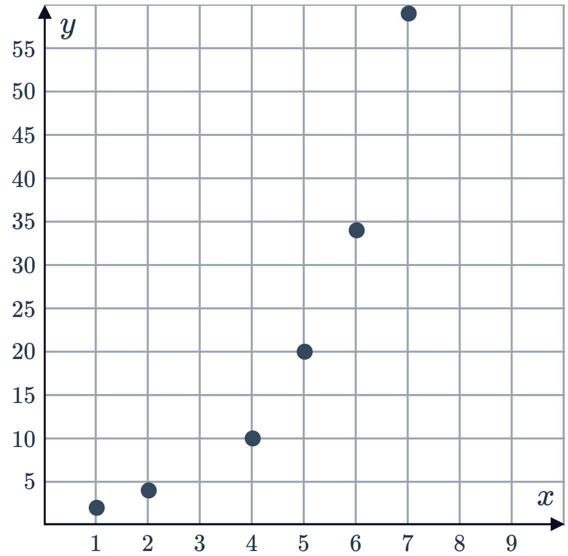
Spearman's rank correlation coeffic

1 For each of the following graphs, determine whether Pearson's correlation coefficient or Spearman's rank coefficient is more suitable for the data set. Explain your choice.

a



b



2 For each of the following sets of data:

i Add ranks for each value to the table.

ii Calculate Spearman's rank coefficient. Round your answer to two decimal places.

iii Describe the correlation between the two variables.

a

x	3	4	5	6	7	8	9
y	7	7.4	7.88	7.64	7.72	8.2	7.32

b

x	2	6	7	14	17	22
y	2.0	2.4	2.0	0.6	0.9	0.2

c

x	1	4	7	10	13	16	19
y	-1	-1.7	-1.7	-1.84	-1.91	-1.98	-2.05

d

x	4	6	8	13	17	21
y	0.4	0.9	0.6	1.7	2.4	1.7

3 The following table displays results from an experiment:

x	1	4	5	8	9	11	13	15	18	19
y	2	4	6	8	12	24	30	46	52	64

a Calculate Pearson's correlation coefficient. Round your answer to three siginificant figures.

b Calculate Spearman's rank coefficient. Round your answer to three siginificant figures.

c Describe the correlation between the variables.

Applications

4 A study was conducted to find the relationship between the age at which a child first speaks and their level of intelligence as teenagers. The following table shows the ages at which some teenagers first spoke, and their results in an aptitude test:

Age when first spoke, x	14	27	9	9	16	21	17	10	7	19
Aptitude test results, y	96	69	84	90	101	87	92	99	104	93

a Construct a scatter plot for the data.

b Add the ranks for each variable to the table of values.

c Calculate Spearman's rank coefficient, to three significant figures.

5 A student was performing an experiment to study the relationship between the current and voltage through a resistor. He noted his results in the following table:

Current, c	1	2	3	4	5	6	7	8	9	10
Voltage, v	1	6	3	10	13	10	16	15	21	17

a Construct a scatter plot for the data.

b Add the ranks for each variable to the table of values.

c Calculate Spearman's rank coefficient, to three significant figures.

6 The following table shows the time taken [redacted]sh a lap on a race track for various average speeds:

Speed, s	20	25	30	35	40	45	50	55	60	65
Time. t	85	87	75	82	69	73	60	57	45	49

a Construct a scatter plot for the data.

b Calculate Spearman's rank coefficient, to three significant figures.

c Describe the correlation between speed and time for this data.

7 Sean is a hotdog vendor. He records the maximum temperature of the day and the number of hotdogs sold. The results are displayed in the following table:

Maximum temperature $(^\circ\text{C}), x$	30	34	33	35	33	28	27	31	37	29
Number of hotdogs, y	18	38	26	40	24	8	20	35	43	38

a Construct a scatter plot to represent the data in the table.

b Calculate Spearman's rank correlation coefficient to two decimal places.

c As the temperature increases, describe what happens to the sales.

8 The table lists the time taken to sprint 400 metres by runners who all ran in different temperatures as part of a study:

Temperature $(^\circ\text{C})$	Time (sec)
5	60
2	67
10	48
8	69
1	65
7	49
6	57
4	53
3	59
9	52

a Construct a scatter plot to represent the data in the table.

b How many runners were tested in the study?

c Calculate Spearman's rank correlation coefficient, to three significant figures.

d Describe the correlation between temperature and sprint time for the data.

e Which data point represents an outlier?

f Recalculate Spearman's rank correlation coefficient for this data, excluding the outlier. Round your answer to three significant figures.

9 The marks of 12 students in Maths and Sport were recorded in the following table:

a Construct a scatter plot for the students' mark in Maths vs their mark in Sport.

b Describe the correlation between students' marks in Maths and marks in Sport.

c Which student's scores appear to represent an outlier?

d Calculate Spearman's rank correlation coefficient for this data, excluding the outlier. Round your answer to three significant figures.

Student	Mark in Maths	Mark in Sport
1	63	44
2	92	74
3	60	52
4	79	70
5	88	67
6	81	60
7	61	73
8	91	86
9	72	84
10	42	93
11	66	57
12	92	92

10 The following table shows the average IQ of a random group of people against their height:

Height (cm)	140	145	150	155	160	165	170	175	180	185
IQ	103	95	98	111	85	89	108	145	110	93

a Construct a scatter plot for this data.

b Describe the relationship between IQ and height.

c How tall is the person who appears to be an outlier?

d Calculate Spearman's rank correlation coefficient for this data, excluding the outlier. Round your answer to three significant figures.